Joe Baker

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Profile

I am a Software Engineer, currently working at Indra MicroNav in Maintenance, patching bugs and fixing issues in their C++ Air Traffic Controller Training Simulation codebase. I Graduated from Bournemouth University last year with first class honours. I am looking to further develop my skills, and contribute to meaningful projects, while expanding my experience with different technologies.

Previous experience includes maintaining a large legacy C++ codebase, and developing supporting tools in Python and C#. Strong foundation in object-oriented design, testing, and debugging of complex, safety-critical systems. With Uni and personal projects, also have experience across Java, Kotlin, and Web technologies, along with my current personal project involving embedded programming on an ESP32.

Experience

Software Engineer @ Indra MicroNav (2023-Present)

Software Engineer: 2026-Present
Graduate Software Engineer: 2025-2026
Software Engineering Placement: 2023-2024

- Investigated and fixed bugs and errors, ranging from minor to critical, across the companies C++ Air Traffic Control simulation software suite within the Maintenance team.
- Currently the maintainer of the airport map builder, a tool used in every customers system, across over 50 countries, with me being the primary source of support for the tool.
- Designed and developed an internal automated testing tool in Python to during the creation and execution of nightly regression tests with Jenkins while previously on the Validation team.
- Worked with complex, safety-critical simulation software, following established development, testing, and review processes.
- Awarded *Highly Commended* by Bournemouth University for outstanding performance and contribution during industrial placement.

Education

Bournemouth University (2021-2025)

BSc (Hons) Software Engineering. Achieving first class honours.

- Dissertation & Final Year Project (2025): using Kotlin and Python, developed a Pedestrian Navigation App which focused on the safety of provided routes, receiving a 80% grade.
- Ubiquitous Computing (2024): using React Native, developed a QR/Barcode Scanner App receiving 95% and a Surf Forecast app receiving 90%.
- Application Programming (2023): using Kotlin, developed a Battleships App, getting a 92% grade.
- Web Programming (2022): using PHP and JavaScript, developed an eCommerce site, with a 91% grade.
- Application of Programming Principles (2022): using Python and JavaScript, created an eBook library website with REST API, receiving a 100% grade.

Brockenhurst College (2019-2021)

BTEC Level 3 Extended Diploma in IT at grade D*D*D* (Triple Distinction Star)

Technical Skills

- **Programming Languages:** Python, C++, Java, Kotlin, HTML, JavaScript, C#, PHP, and SQL
- **Design & Architecture:** OOP (Object Oriented Programming), UML (Unified Modeling Language), REST API (Representational State Transfer API), MVC (Model View Controller), EDA (Event-Driven Architecture)
- **Platforms:** Android, Linux, Windows, and Embedded Systems
- **Software:** Docker, Git, SVN, Jenkins, MS Office, VirtualBox, and various IDEs including Android Studio, Embarcadero Builder, Visual Studio, and VS Code

Professional Skills

- **Problem Solving:** Diagnosed and resolved critical C++ software issues and bugs, improving system usability and stability.
- **Adaptability:** Quickly understood and became productive across large unfamiliar codebases, tools, and technologies, learning approaches to contribute effectively with changing roles and projects.
- **Teamwork:** Collaborated with others both at MicroNav and in university group projects, to share responsibilities and workloads, support teammates, and resolve issues, to achieve success.
- **Communication:** Presented updates and technical documentation, to both technical and non-technical audiences, while at MicroNav and during university projects.
- **Time Management:** Balanced multiple tasks and deadlines while consistently meeting deadlines and staying on schedule.

References

Available upon request